

Project Title: EduTrack – Smart Student Management System

Submitted By: Kasa Mounika

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Technology Stack: React.js, Node.js, Express.js, MongoDB Atlas

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Abstract

EduTrack is a Smart Student Management System developed to simplify academic administration and student record management. The application provides a centralized platform for managing students, teachers, parents, courses, departments, sections, and academic information. The project was implemented progressively across three levels, starting from REST API development and evolving into a complete academic management platform with authentication, role-based access control, and advanced academic workflows.

Introduction

Educational institutions require efficient systems to manage student records, academic information, and communication among administrators, teachers, students, and parents. Traditional manual methods are time-consuming and prone to errors. EduTrack was developed to address these challenges by providing a secure, scalable, and user-friendly web-based management system.

Project Objectives

The main objectives of the project are:

- Develop a REST API using Node.js and Express.js.
- Implement CRUD operations for student management.
- Integrate MongoDB Atlas for database storage.
- Build a responsive frontend interface.
- Implement authentication and authorization.
- Provide role-based access for different users.
- Create a centralized academic management system.
- Deploy the application on cloud infrastructure.

Technologies Used

Frontend:

- React.js
- Vite
- Tailwind CSS
- Axios
- React Router

Backend:

- Node.js
- Express.js
- JWT Authentication
- GraphQL
- Socket.io

Database:

- MongoDB Atlas
- Mongoose

Deployment:

- Render
- MongoDB Atlas

System Architecture

The system follows a client-server architecture.

Frontend:

The React-based frontend provides an interactive user interface for all users.

Backend:

The Express.js server handles API requests, authentication, business logic, and database operations.

Database:

MongoDB Atlas stores all application data including users, students, teachers, parents, courses, and academic records.

Deployment:

The frontend and backend are deployed on Render while MongoDB Atlas hosts the database.

Level 1 Implementation

Level 1 focused on REST API development and CRUD operations.

Features Implemented:

- Express server setup
- MongoDB integration
- Student CRUD APIs
- Error handling
- HTML, CSS, and JavaScript frontend
- Dynamic student data display

CRUD Operations:

- Create Student
- Read Student Information
- Update Student Information
- Delete Student Records

The image displays two screenshots of the EduTrack Smart Student Management System interface. The top screenshot shows the 'Add Student' form and the 'All Students' directory. The bottom screenshot shows the same interface but with the 'All Students' directory displaying student profiles with names and IDs.

EduTrack SMART STUDENT MANAGEMENT SYSTEM API Connected

STUDENT FORM Add Student

Profile Photo: Choose File No file chosen

Student ID: EDU-1001 Gender: Select gender

Name: Enter full name

First Name: First name Last Name: Last name

Date of Birth: dd-mm-yyyy

Email: student@example.com

DIRECTORY All Students 4 students

Search students: Search by name, email, or course Refresh

	COURSE	AGE	ACTIONS
ukku47@gmail.com	6a1ad8b782a03152d7170dbc	20	View Profile Edit Delete
566762@gmail.com	6a1ad8b782a03152d7170dbc	20	View Profile Edit Delete
@edutrack.com	6a1ad8b782a03152d7170dbc	-	View Profile Edit Delete
@edutrack.com	6a1ad8b782a03152d7170dbc	-	View Profile Edit Delete

EduTrack SMART STUDENT MANAGEMENT SYSTEM API Connected

STUDENT FORM Add Student

Profile Photo: Choose File No file chosen

Student ID: EDU-1001 Gender: Select gender

Name: Enter full name

First Name: First name Last Name: Last name

Date of Birth: dd-mm-yyyy

Email: student@example.com

DIRECTORY All Students 4 students

Search students: Search by name, email, or course Refresh

NAME	EMAIL	COURSE	AGE	ACTIONS
Charan EDU-1234	charanukku47@gmail.com	6a1ad8b782a03152d7170dbc	20	View Profile
Puppy EDU-123	mounika566762@gmail.com	6a1ad8b782a03152d7170dbc	20	View Profile
Mounika EDU-1002	mounika@edutrack.com	6a1ad8b782a03152d7170dbc	-	View Profile
Charan EDU-1001	charan@edutrack.com	6a1ad8b782a03152d7170dbc	-	View Profile

Level 2 Implementation

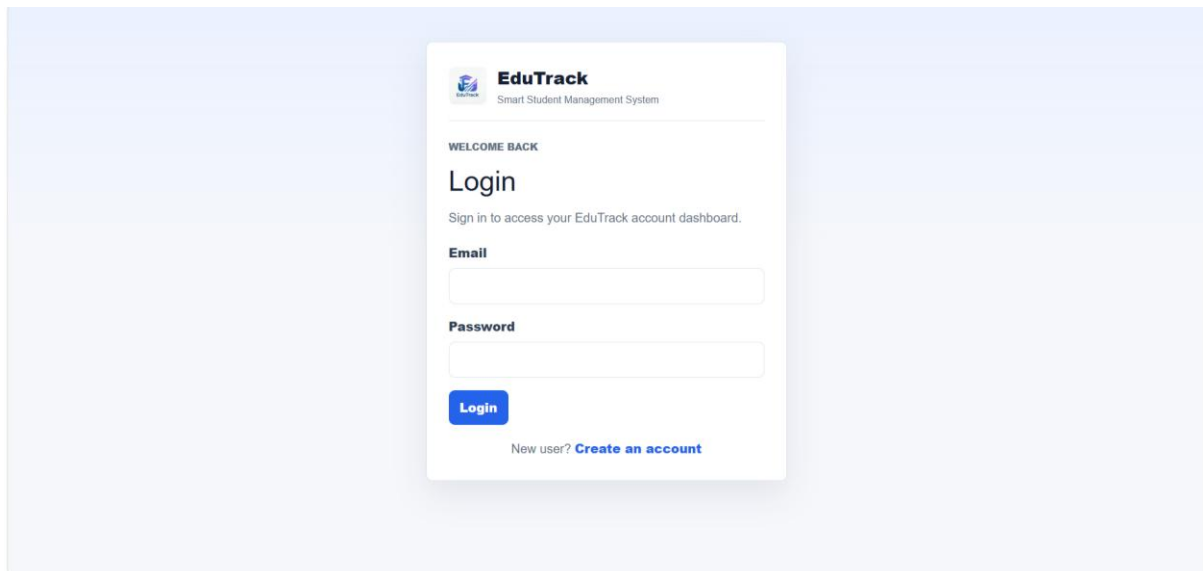
Level 2 expanded the project into a complete management system.

Features Implemented:

- User Authentication
- JWT Authorization
- Dashboard
- Student Management
- Teacher Management
- Parent Management
- Course Management
- Profile Management

Modules:

- Students
- Teachers
- Parents
- Courses
- Dashboard
- Profile



 **EduTrack**
Smart Student Management System

CREATE ACCOUNT

Register

Choose Admin for full management access or Student for profile-only access.

Name

Email

Password

Access Type

Student

Parent Name

 **EduTrack**
Smart Student Management System

Dashboard

Admin Principal

Logout

MENU

Dashboard

Students

Teachers

Parents

Courses

Profile

Admin

ADMIN WORKSPACE

EduTrack Dashboard

Role-based academic overview and student records system.

Students

4

Teachers

2

Parents

3

Courses

2

Recent Students

Manage

C

Charan

charanukku47@gmail.com

P

Puppy

mounika566762@gmail.com

M

Mounika

mounika@edutrack.com

 **EduTrack**
Smart Student Management System

Students

Mr. Ravi

Logout

MENU

Dashboard

Students

Profile

Teacher

DIRECTORY

Students

3 students

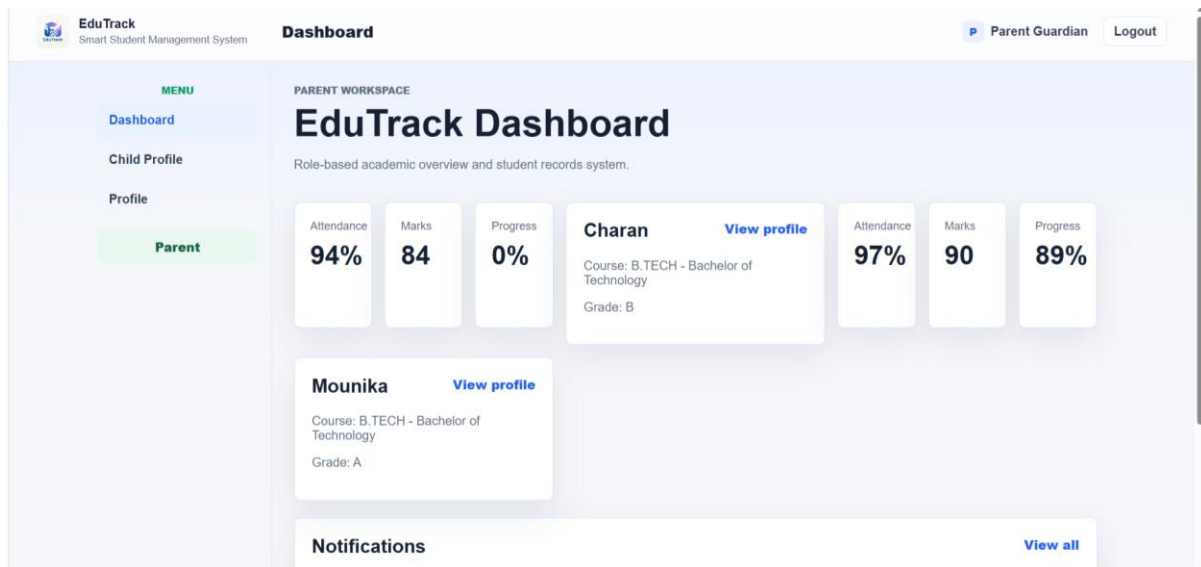
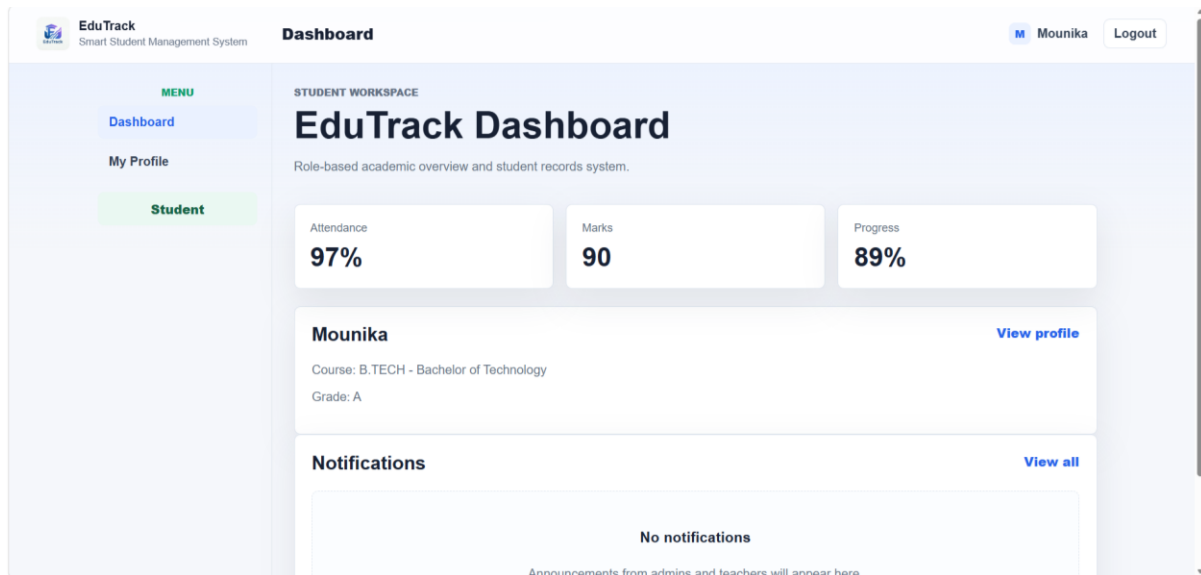
Search, view, edit, and remove student records.

Search students

Search by name, email, or course

NAME	EMAIL	COURSE	AGE	ATTENDANCE	MARKS	GRADE	ACTIONS
Charan EDU-1234	charanukku47@gmail.com	B.TECH - Bachelor of Technology	20	0%	0	-	View Update Progres
Mounika EDU-1002	mounika@edutrack.com	B.TECH - Bachelor of Technology	-	97%	90	-	View Update Progres
Charan EDU-1001	charan@edutrack.com	B.TECH - Bachelor of Technology	-	94%	84	-	View Update Progres

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Level 3 Implementation

Level 3 introduced advanced academic management features.

Features Implemented:

- Department Management
- Section Management
- Subject Management
- Teacher Assignment
- Parent-Student Mapping
- Notifications
- GraphQL Integration
- Socket.io Integration
- Advanced Academic Management

EduTrack

Smart Student Management System

Dashboard

My Subjects

My Sections

Students

Attendance

Marks

Announcements

Profile

M

Mr. Ravi

ravi@edutrack.com

Teacher

Dashboard

TEACHER DASHBOARD

Welcome, Mr. Ravi

Track assigned sections, update academic records, and follow classroom activity.

Students

3

Visible learner records

Courses

2

Active academic programs

Attendance

64%

Average visible attendance

Average Score

58

Across visible learners

Recent Academic Activity

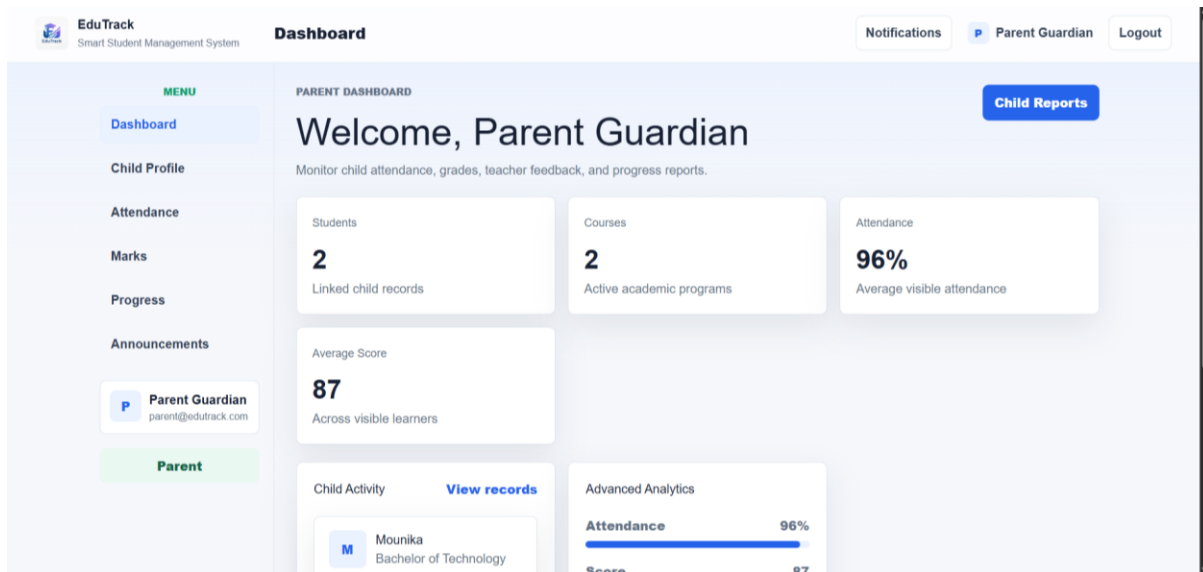
View records

Mounika

Advanced Analytics

Attendance

64%



Database Design

The MongoDB database consists of multiple collections:

- Users
- Students
- Teachers
- Parents
- Courses
- Departments
- Sections
- Subjects
- Notifications

Relationships are maintained between students, parents, teachers, departments, sections, and courses to ensure data consistency.

Modules Description

Authentication Module

Provides secure login and registration functionality using JWT authentication.

Student Management Module

Allows administrators to add, edit, view, and delete student information.

Teacher Management Module

Manages teacher records, assignments, and profile information.

Parent Management Module

Maintains parent records and links parents with students.

Course Management Module

Handles course creation, modification, and management.

Academic Management Module

Manages departments, sections, subjects, and teacher assignments.

Profile Management Module

Allows users to update personal information and profile pictures.

System Workflow

Administrator:

- Creates courses
- Creates departments
- Creates sections
- Adds students
- Adds teachers
- Adds parents
- Assigns teachers

Teacher:

- Views assigned students
- Manages academic activities

Parent:

- Monitors student information

Student:

- Views personal profile and academic information

Deployment Details

Frontend Deployment:

<https://edutrackf.onrender.com>

Backend Deployment:

<https://edutrack-c38t.onrender.com>

Database:

MongoDB Atlas

Testing and Validation

Testing was conducted to verify:

- User Authentication
- CRUD Operations
- Data Validation
- Role-Based Access Control
- Database Connectivity
- Deployment Functionality

All major modules were tested successfully.

Challenges Faced

- MongoDB Atlas connectivity configuration.
- CORS configuration during deployment.
- Environment variable management.
- Frontend and backend integration.
- Role-based access implementation.

Results and Achievements

- Successfully developed a complete MERN-stack application.
- Implemented secure authentication and authorization.
- Built a scalable academic management platform.
- Successfully deployed the application on cloud infrastructure.
- Implemented advanced academic workflow management.

Conclusion

The EduTrack – Smart Student Management System was successfully designed and developed as a full-stack web application to simplify academic administration and student record management. The project demonstrates the practical implementation of modern web technologies including React.js, Node.js, Express.js, MongoDB Atlas, JWT Authentication, GraphQL, and Socket.io.

The project was developed progressively through three levels. Level 1 focused on REST API development and CRUD operations, Level 2 introduced authentication and role-based management features, and Level 3 expanded the system into a comprehensive academic management platform with advanced functionality.

The system provides an efficient solution for managing students, teachers, parents, courses, departments, sections, subjects, and academic workflows. It offers a secure, scalable, and user-friendly environment that improves data organization, accessibility, and overall administrative efficiency.

Through this project, valuable experience was gained in full-stack development, database management, API integration, authentication, deployment, and real-world software engineering practices. EduTrack successfully meets the project objectives and serves as a strong foundation for future enhancements such as attendance management, examination systems, timetable scheduling, and mobile application support.